

### Calculus 1 sec 1: Assignment 3, due Thursday Mar 26, 2025, 11:59pm

Submit the completed assignment to <https://www.gradescope.com>.

Only hand-written solutions are accepted.

- This assignment is worth a total of 15 points + 4 bonus points, and accounts for about 3% of your final grade. The late submission deadline (no penalty) is Friday, Mar 27 11:59 pm. Submissions after the late due date will NOT be accepted unless you have a good excuse.
- Collaboration is allowed, but you must write up your solutions independently and in your own words. Copying from others, publications, or websites (even with minor changes) will be treated as plagiarism.
- Neatness, clarity, and correct notation matter! Show all steps, use clear explanations, and write in complete sentences where needed. Graders will only evaluate what you write, so unclear or incomplete work will affect your grade, even with the correct answer.
- Attempt all questions (and parts), and clearly indicate which question you are answering. Answer each question on a separate sheet, and please **match your answers with the correct questions on Gradescope**.

1 (a) [1 pt] Suppose that a differentiable function  $f$  satisfies the property that  $|f(x)| \leq x^2$  for all  $x$ . Show that  $f(0) = 0$  and  $f'(0) = 0$ . (Use the definition of the derivative as a limit.)

(b) [2 pt] Determine whether or not the following functions are differentiable at  $x = 0$ . Fully justify your answer, using the definition of the derivative as a limit.

$$(i) \quad g(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & x \neq 0, \\ 0 & x = 0. \end{cases} \quad (ii) \quad h(x) = \begin{cases} x \cos(\frac{1}{x}) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

2. [2 pt] Find the limit, showing all the steps and using the fact that  $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ ,

$$(i) \lim_{x \rightarrow 0} \frac{\sin(x^2)}{\sin^2(3x)} \quad (ii) \lim_{x \rightarrow 1} \frac{(1 - \cos(\pi x)) \sin(x - 1)}{x^2 - 1}$$

3. [2 pt] Find constants  $a, b$  such that the function  $y = a \sin(x) + b \cos(x)$  is a solution to the equation  $3y'' - y' + 2y = \sin(x) + 2\cos(x)$ .

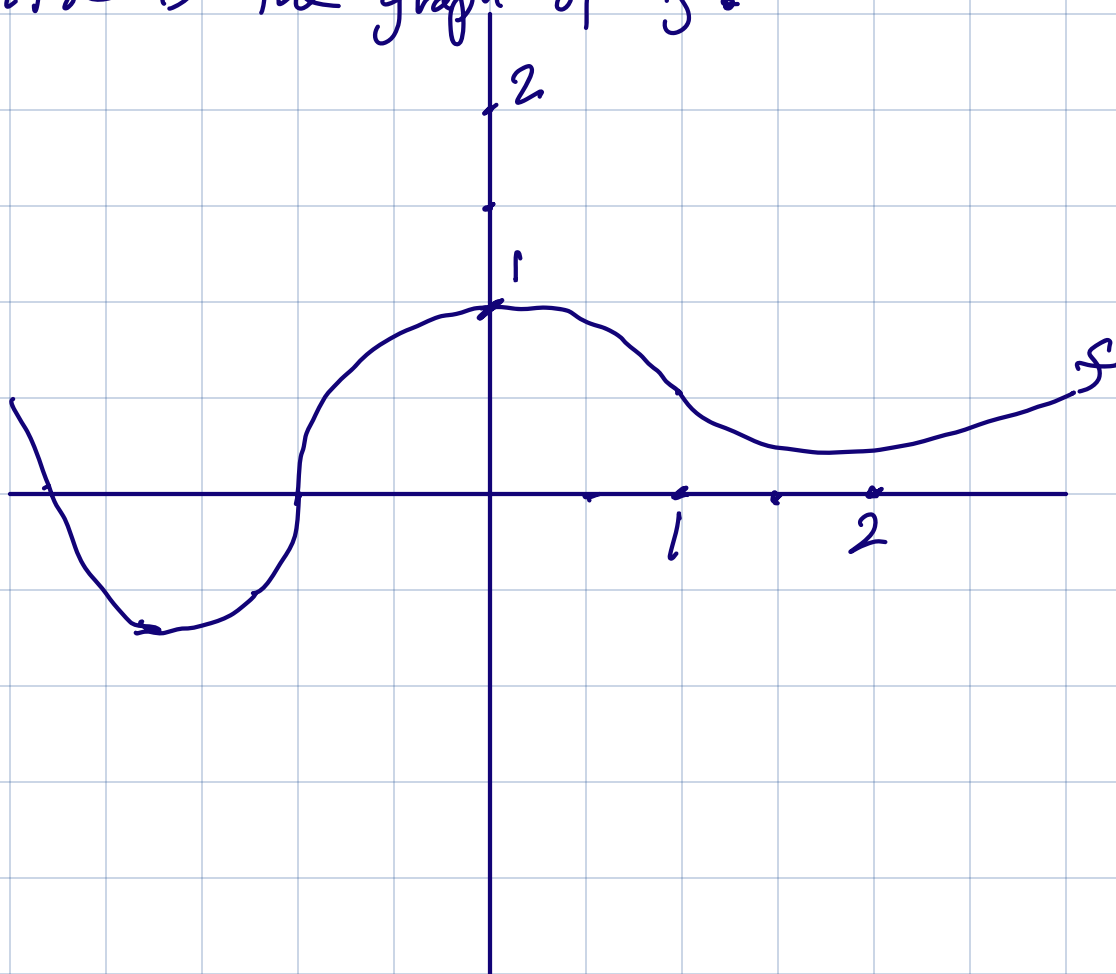
(b) [3 pt] Let  $x(t)$  be a function such that  $e^{2x} \cos(t) = xt + 2e$ . Find an expression for  $\frac{dx}{dt}$  and evaluate both  $x$  and  $\frac{dx}{dt}$  at  $t = 0$ . *Show all the steps*

4. (i) [1 pt] For which constants  $c$  does the equation  $e^{\cos(x)} = c$  have at least one solution?

(ii) [2 pt] Let  $f(x) = e^{\cos(x)}$ . Calculate  $f'$  and  $f''$ .

(iii) [2 pt] With  $f$  as in (ii), show that the equation  $f''(x) = 0$  has a solution for some  $x$  between  $\pi/4$  and  $\pi/3$ . Explain your reasoning.

Here is the graph of  $f$ .



a) (1 pt) Estimate  $f'(x)$  at  $x = -2, 0, 1$ .

b) (1 pt) What is  $\lim_{x \rightarrow -1} f'(x)$ ? Does it exist?

c) 2pt Add graph  $f'$  to the above figure, being as accurate as you can.